

Sample Problems
Lesson: Random Variables
Second Semester, SY 2025 - 2026

General Instructions: Do as instructed.

1. Classify the following variables as discrete or continuous.
 - (a) X : Number of requests received by a server in 1 minute
 - (b) Y : CPU temperature (in °C)
 - (c) Z : Number of users currently connected
 - (d) W : Time (in seconds) it takes to process a request
 - (e) A : Bandwidth usage (in Mbps)
2. A system generates a 3-character password using the letters A, B, C and the digits 1 and 2 without restrictions. Let X be a random variable representing the number of digits in the generated password.
 - (a) Determine the probability distribution function (PMF) of X .
 - (b) Construct the cumulative distribution function (CDF) of X .
3. The cumulative distribution function of T is given by

$$F(t) = \begin{cases} 0, & t < 0 \\ \frac{1}{8}, & 0 \leq t < 5 \\ \frac{1}{4}, & 5 \leq t < 7 \\ \frac{1}{2}, & 7 \leq t < 9 \\ \frac{7}{8}, & 9 \leq t < 10 \\ 1, & t \geq 10 \end{cases}$$

Determine the following:

- (a) $P(T = 5)$
 - (b) $P(T = 8)$
 - (c) $P(T > 7)$
 - (d) $P(5 \leq T < 10)$
 - (e) $P(T \geq 5 \mid T > 0)$
4. Determine the value of c such that $f(x) = c(x^2 + 4)$ serves as a probability distribution of the discrete random variable X , for values 0, 1, 2, and 3.
5. A system administrator is modeling the server response time (in seconds) for a lightweight API. Let the continuous random variable X represent the response time, and suppose it follows the probability density function:

$$f(x) = \begin{cases} c(x^2 + 1), & 0 \leq x < 2, \\ 0, & \text{elsewhere} \end{cases}$$

Where c is a constant that makes $f(x)$ a valid probability density function.

- (a) Determine the value of c such that $f(x)$ is a valid probability density function.
 - (b) Find the probability that the response time is between $\frac{1}{2}$ and $\frac{3}{2}$ seconds.
 - (c) Construct the cumulative distribution function (CDF) of X .
 - (d) Use $F(x)$ to determine the probability that the response time is between $\frac{1}{2}$ and $\frac{3}{2}$ seconds.

6. Determine the value of c such that the function

$$f(x, y) = cxy, \quad \text{for } x = 1, 2, 3 \text{ and } y = 1, 2, 3,$$

represents a joint probability distribution of the random variables X and Y .

7. A server log contains 8 file requests, consisting of 3 video files, 2 image files, and 3 document files. A random sample of 4 file requests is selected. Let X be the number of video files in the sample, and Y be the number of image files in the sample.

- Find the joint probability distribution of X and Y .
- Determine the marginal distribution of X and Y .
- Determine the conditional distribution of X when $Y = 1$.
- Determine and show if X and Y are statistically independent.
- Compute the following probabilities:
 - $P(X \leq 2, Y = 1)$
 - $P(X > 2, Y \leq 1)$
 - $P(X > Y)$
 - $P(X + Y = 4)$
 - $P(X = 2 \mid Y = 1)$

8. A web server records the response time for user requests in seconds. Let X be the time (in seconds) to process a video file request and Y be the time (in seconds) to process an image file request. Suppose the joint density function of X and Y is

$$f(x, y) = \begin{cases} x + y, & 0 \leq x \leq 1, 0 \leq y \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

- Verify that $f(x, y)$ is a valid joint density function.
- Find the marginal densities $g(x)$, $h(y)$, and the conditional density $f(y \mid x)$.
- Determine and show whether X and Y are statistically independent.
- Compute the following probabilities:
 - $P(X \leq \frac{1}{2}, Y \leq \frac{1}{2})$
 - $P(X > \frac{1}{2}, Y \leq \frac{1}{2})$
 - $P(Y \leq \frac{1}{2} \mid X = \frac{3}{10})$

9. A server handles three types of tasks in one minute - video, image, and document tasks. Let X_1 be the number of video tasks, X_2 be the number of image tasks, and X_3 be the number of document tasks. The probability mass functions of X_1 , X_2 , and X_3 are given below. Assuming independence, determine $P(X_1 < 2, 1 < X_2 < 3, X_3 > 2)$.

$$P(X_1 = x_1) = \frac{1}{3}, \quad x_1 = 0, 1, 2 \quad P(X_2 = x_2) = \frac{1}{4}, \quad x_2 = 0, 1, 2, 3 \quad P(X_3 = x_3) = \frac{1}{4}, \quad x_3 = 0, 1, 2, 3$$

10. A server measures processing times (in seconds) for three types of tasks - video, image, and document tasks. Let X_1 be the processing time of video tasks, X_2 be the processing time of image tasks, and X_3 be the processing time of document tasks. The probability density functions of X_1 , X_2 , and X_3 are given below. Assuming independence, determine $P(X_1 < \frac{1}{2}, \frac{1}{5} < X_2 < \frac{3}{5}, X_3 > \frac{1}{2})$.

$$f(x_1) = 2x, \quad 0 \leq x \leq 1 \quad f(x_2) = 3x^2, \quad 0 \leq x \leq 1 \quad f(x_3) = 4(1 - x)^3, \quad 0 \leq x \leq 1$$